

Superdeterminism lives below every resolution; a floor gives it a lifetime

Result, 2026-09-29 (Claude; written derivation, not yet refereed). Classical mechanics is deterministic, so the settings of a Bell test are functions of the initial data; the statistical independence of settings and hidden variables that Bell's theorem needs is an assumption of typicality on those data, taken with the Liouville measure. Superdeterminism is the choice of atypical data. The proposition below shows where such data must hide when the setting device is chaotic: in structure of the initial ensemble finer than any fixed phase-space resolution. For a preparation ensemble resolved at scale δ , the covariance between a hidden variable and a setting chosen n steps later obeys

$$|\text{Cov}_\rho(f, g \circ T^n)| \leq 2C_T K \|f\|_{\text{Lip}} \|g\|_{\text{Lip}} \delta^{-(d+1)} \theta^n,$$

so the correlation is below η after

$$n_*(\delta) = \frac{(d+1) \ln(1/\delta) + \ln(2C_T K \|f\|_{\text{Lip}} \|g\|_{\text{Lip}} / \eta)}{\ln(1/\theta)}$$

steps. Classical mechanics puts no lower bound on δ , and $n_* \rightarrow \infty$: a conspiracy can be stored in ever finer structure for as long as needed. If a floor h on phase-space area per canonical pair constrains the physical state, then $\ln(1/\delta) = \frac{1}{2} \ln(A_0/h)$ with A_0 the action scale of the device, and the conspiracy has a finite lifetime $n_* \approx \frac{d+1}{2 \ln(1/\theta)} \ln(A_0/h)$, the logarithmic structure of the Ehrenfest time of quantum chaos (Berman and Zaslavsky 1978; metadata). With the floor, the dependence between hidden variable and setting decays exponentially after an onset of order n_* , also for setting and hidden-variable events (Remark in §1), and relaxed Bell inequalities then bound local deterministic models that respect the floor. The limit $h \rightarrow 0$ that Newton takes is exactly what reopens the loophole.

The user's remark that prompted this (2026-09-28/29): classical mechanics is accidentally superdeterministic too, and the programme holds that it has a mistake somewhere. The link is that typicality rests on the Liouville measure, whose unit is an action per degree of freedom, the same object whose unlimited refinement the fifth-postulate note identifies as Newton's joint determinacy.

1. Setting and the proposition

Let M be a compact d -dimensional manifold with normalized volume μ , and T a C^2 volume-preserving Anosov diffeomorphism, the setting device. Such maps are mixing with exponential decay of correlations for Hölder, hence Lipschitz, observables (Bowen 1975; metadata): with $\|F\|_{\text{Lip}} = \sup |F| + \text{Lip}(F)$,

$$\left| \int F(G \circ T^n) d\mu - \int F d\mu \int G d\mu \right| \leq C_T \|F\|_{\text{Lip}} \|G\|_{\text{Lip}} \theta^n, \quad \theta < 1.$$

A preparation ensemble is a probability density ρ on M ; the hidden variable of the measured system is $\lambda = f(x)$ and the setting is $s = g(T^n x)$, with f, g Lipschitz. The ensemble is **resolved at scale** δ if $\|\rho\|_{\text{Lip}} \leq K \delta^{-(d+1)}$; for example any probability measure smoothed by a mollifier of width δ , whose density has height of order δ^{-d} and slope of order $\delta^{-(d+1)}$.

Proposition. For an ensemble resolved at scale δ , $|\text{Cov}_\rho(f, g \circ T^n)| \leq 2C_T K \|f\|_{\text{Lip}} \|g\|_{\text{Lip}} \delta^{-(d+1)} \theta^n$.

Proof. Apply the decay estimate twice, with $F = f\rho$ and with $F = \rho$, using $\int \rho d\mu = 1$: $\int f\rho(g \circ T^n) = \int f\rho \int g + E_1$ and $\int \rho(g \circ T^n) = \int g + E_2$. Then $\text{Cov}_\rho = E_1 - E_2 \int f\rho$, and $\|f\rho\|_{\text{Lip}} \leq \|f\|_{\text{Lip}} \|\rho\|_{\text{Lip}}$, $|\int f\rho| \leq \|f\|_{\text{Lip}}$ give the bound. $\square$

Remark (from covariances to measurement dependence). Bell's theorem uses the laws of λ conditional on the settings, and its relaxed forms measure their dependence in total

variation (Hall 2010; Barrett and Gisin 2011; Crossref metadata verified 2026-09-29). Hall’s deterministic model of the singlet uses only a partial measurement dependence, so this conversion carries weight. Let the setting and hidden-variable events $A, B \subset M$ have piecewise smooth boundaries whose w -neighbourhoods have μ -measure at most $c_A w$ and $c_B w$. Replace $1_A, 1_B$ by functions with values in $[0, 1]$ and Lipschitz constant $1/w$ that differ from them only there, so $\|\cdot\|_{\text{Lip}} \leq 2/w$ for $w \leq 1$. Since T preserves μ , each boundary layer costs at most $\sup \rho c w \leq K \delta^{-(d+1)} c w$ under ρ , in $E_\rho[1_B 1_A \circ T^n]$ and in the product of the means. Hence

$$|\text{Cov}_\rho(1_B, 1_A \circ T^n)| \leq K \delta^{-(d+1)} [8C_T \theta^n w^{-2} + 2(c_A + c_B)w],$$

and $w = \theta^{n/3}$ gives $K \delta^{-(d+1)} (8C_T + 2c_A + 2c_B) \theta^{n/3}$. For finitely many settings and a finite partition of hidden-variable events, summing gives a total-variation dependence with the same decay. The onset for events is three times the Lipschitz n_* , with the same logarithm of A_0/h .

2. What it says, and about whom

- **Classical mechanics.** Nothing in the laws fixes δ ; an ensemble can be a point mass. Independence is therefore an assumption about the data, and the superdeterministic option is a choice of data with structure below every scale. This is the precise sense of “accidentally superdeterministic”.
- **A floor.** The proposition needs the floor to constrain the physical state (its phase-space density), beyond constraining records. Under that premise, correlations of a chaotic setting device with a hidden variable decay after a time logarithmic in A_0/h , the Ehrenfest-type scale at which classical and quantum chaos part.
- **’t Hooft’s automaton** lies outside the proposition. Its ontic states are exact cells and its correlations are carried by exact labels, with no Lipschitz structure to smooth; a floor on phase-space density says nothing about them (discrete-substrate note).
- **Palmer’s invariant set postulate** places states on a fractal invariant set and counterfactual settings off it (Palmer 2009; read at the level of its title and our memory of the framework). A fractal support is structure at all scales, which a floor in the proposition’s sense smears out; the two premises are in direct conflict. The superdeterminist position is surveyed in Hossenfelder and Palmer (2020) (metadata).
- **Quantum mechanics** is untouched: its Bell violations come from entangled states, and it assumes independent settings, consistently with the proposition.

3. Consequence for STATE

A small addition to the Newton side: for chaotic setting devices the floor makes Bell’s measurement dependence decay exponentially after an Ehrenfest-type onset, and the continuum limit $h \rightarrow 0$ reopens the loophole. It is a statement about what the floor buys, with the floor assumed; the necessity question is unchanged. To be refereed.